

```
19 cipher_gui.py > ...
20
21 import tkinter as tk
22 from tkinter import ttk
23
24 import cipher_tool as ct
25
26 # -----
27 # Palette -- matches the "cipher Desk" web version (aged paper / ink / stamp)
28 # -----
29 PAPER = "#E6E6D6"
30 PAPER_DEEP = "#E2DBC7"
31 CARD = "#F6F2E7"
32 LINE = "#C9BFA8"
33 INK = "#22283A"
34 INK_SOFT = "#5B6178"
35 STAMP = "#B23A2E"
36 STAMP_SOFT = "#EBC9C2"
37 FIELD = "#4C6B57"
38 FIELD_SOFT = "#0CE5D0"
39
40 FONT_DISPLAY = ("Georgia", 11)
41 FONT_DISPLAY_BOLD = ("Georgia", 11, "bold")
42 FONT_HEAD = ("Georgia", 22, "bold")
43 FONT_SUBHEAD = ("Georgia", 13, "bold")
44 FONT_MONO = ("Consolas", 10)
45 FONT_MONO_SM = ("Consolas", 9)
46
47
48 def clean(text: str) -> str:
49     return ct.clean_text(text)
50
51
52 def guess_key_length_details(ciphertext: str, max_len: int = 20):
53     """Same computation as cipher_tool.guess_key_length, but also returns
54     the per-length (ioc, metric) details so the GUI can chart them."""
55     kasiski_scores = ct.kasiski_key_length_candidates(ciphertext, max_len)
56     details = []
57     best_len, best_metric = 1, -1.0
58     for length in range(1, max_len + 1):
59         ioc = ct.average_ioc_for_key_length(ciphertext, length)
60         closeness = 1.0 - abs(ioc - ct.ENGLISH_IOC) / ct.ENGLISH_IOC
```

```
1 #!/usr/bin/env python3
2 """
3 cipher_gui.py
4 =====
5
6 A Tkinter desktop GUI for the Caesar & Vigenere cipher tool.
7
8 Run it with:
9     python3 cipher_gui.py
10
11 Requires: Python 3 with tkinter (included with most Python installs; on
12 some Linux distros install it separately with "sudo apt install python3-tk").
13
14 This GUI is a thin presentation layer over cipher_tool.py -- all the actual
15 cipher and cryptanalysis logic (Caesar shift, Vigenere, chi-squared
16 frequency scoring, Kasiski examination, Index of Coincidence) lives there
17 and is imported unchanged. cipher_tool.py must be in the same folder as
18 this file.
19 """
20
21 import tkinter as tk
22 from tkinter import ttk
23
24 import cipher_tool as ct
```

Encrypt, decrypt, or break a Caesar or Vigenère cipher using letter-frequency statistics — no key required.

Vigenère cipher

Every letter shifts by the same fixed amount.

Trainer Name GAnESH KUCHe

Shift (0-25)

Decrypt

Shift (0-25) 11

Crack it (no key)

Result

Eclypc Ylxp RLypDS VFnSp

Copy

Letter frequency: your text (red) vs. standard English (tan)



Encrypt, decrypt, or break a Caesar or Vigenère cipher using letter-frequency statistics — no key required.

Vigenère cipher

The keyword repeats, letter by letter, choosing a different shift each time.

Trainer Name GAnESH KUCHe LABMentix
attack at dawn

Keyword

🔒 Encrypt

Transform

Crack it (no key)

How the keyword lines up

LEMONLEMONLEMONLEMONLEMONLEMONLEMONLEMONLEMONLEM
TRAINERNAMEGANESEHKUCHELABMENTIXATTACKATDAWN

Result

Evmwapy Zozp KMrDL WIpSi X0oXizhvi
efhnno mh qlaz

Copy

